IndicSent-M

Multimodal Sentiment Analysis for Indian Languages Using Text and Images

XLM-R + ResNet-50 on Bengali memes (MemoSen) · text vs image vs fusion · explainability

Positive | Neutral | Negative

Why memes need more than text

Motivation and problem statement

- **Multilingual, code-mixed.** Indian social media mixes Bengali and Latin script, romanised Bengali/Hindi, emojis and cultural references.
- **Meaning lives across modalities.** A cheerful caption on a grim picture can be sarcasm; the image alone or the text alone can mislead.
- **Tools lag behind.** Most sentiment systems are text-only and English-centric; Indian languages are lower-resource.
- **Question.** Does fusing caption + image beat either modality alone, and can we show why a model decided?

4,362

labelled Bengali memes used

21%

captions mixing Bengali and Latin script

7%

of memes are neutral: strong imbalance

Research objectives

1

Build three model types

Text-only (XLM-R), image-only (ResNet-50) and multimodal fusion

2

Compare fairly

Same leakage-safe split, same training recipe, bootstrap confidence intervals

3

Two fusion strategies

Feature concatenation baseline vs. bidirectional cross-attention

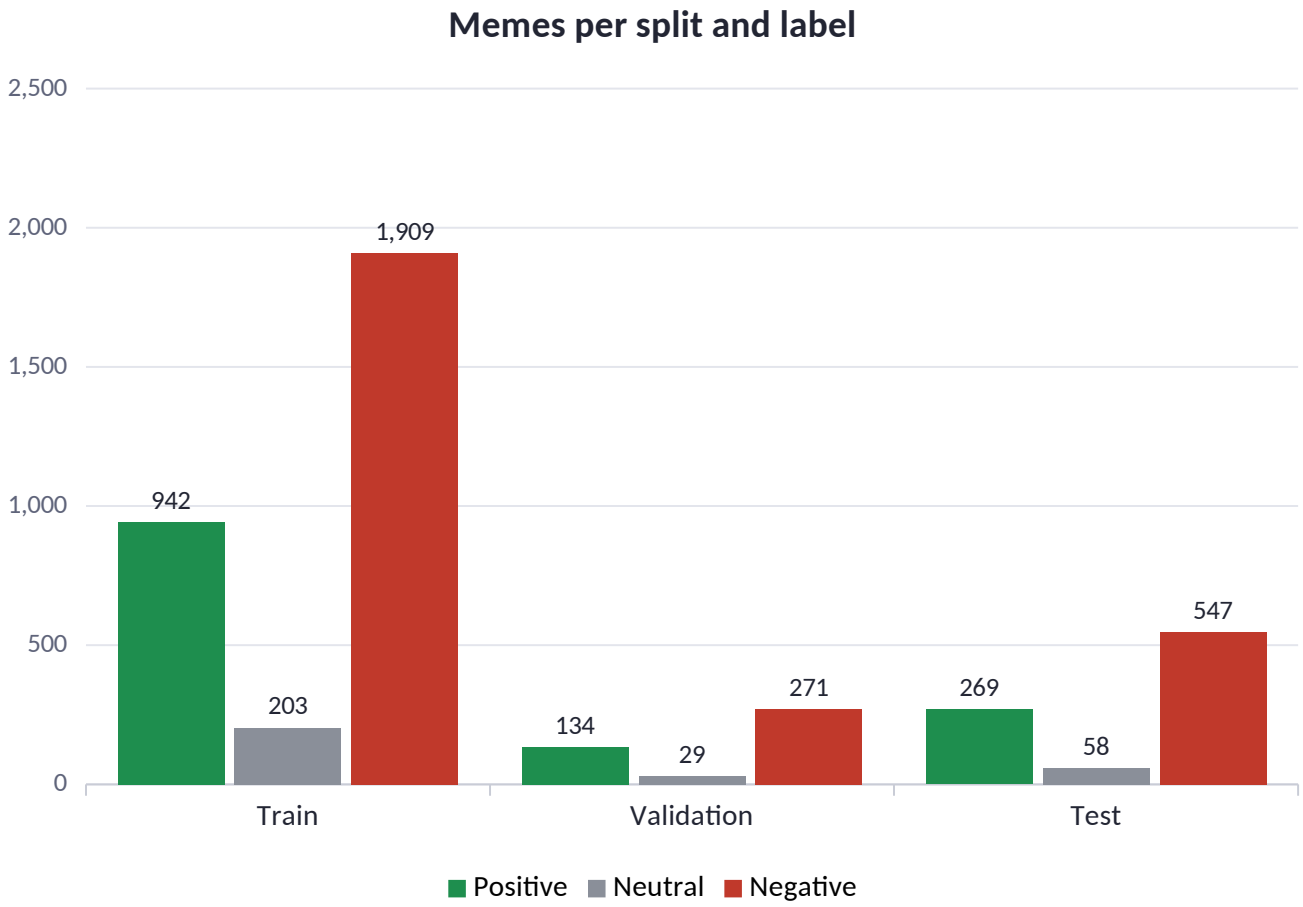
4

Explain predictions

Word occlusion / Integrated Gradients and Grad-CAM, with stated limits

Dataset: MemoSen (Bengali memes, LREC 2022)

4,372 annotations → 4,362 usable after validation & de-duplication



- **Inputs:** meme image + transcribed caption; label positive / neutral / negative
- **Scripts:** Bengali 2,963 · mixed 905 · Latin 494
- **Split:** 70 / 10 / 20, stratified, grouped by near-duplicate image or caption — 0 groups leak
- **Removed:** 6 exact duplicate, 4 invalid image
- **Licence:** none published → research use with citation, not redistributed

Preprocessing pipeline

Conservative by design: sentiment cues and scripts are preserved

1. Load

xlsx captions, image files, labels

2. Validate

decode, size limits, drop missing

3. Normalise

NFC · URL/@USER · keep emoji, negation, ZWJ

4. Script ID

Bengali / Latin / mixed

5. De-duplicate

dHash + text; conflicting labels dropped

6. Group split

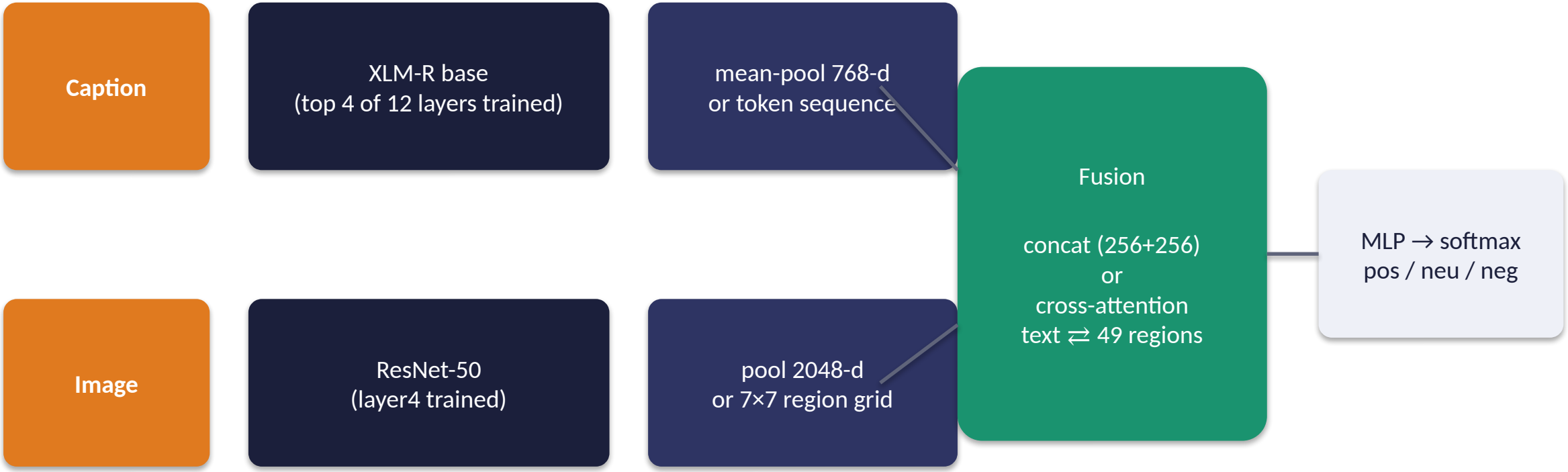
union-find groups, seed 42

Not done on purpose:

lower-casing (XLM-R is cased), emoji/stop-word removal, stemming, transliteration — each would erase sentiment cues or script information. Images cached at 448 px once so training does not decode multi-megapixel files every epoch.

Model architecture

Three model families, one shared training recipe



Text-only and image-only models put a linear head directly on the pooled vector. Multimodal models always use both inputs — forward() refuses a missing modality, and tests check both change the output.

Training and evaluation protocol

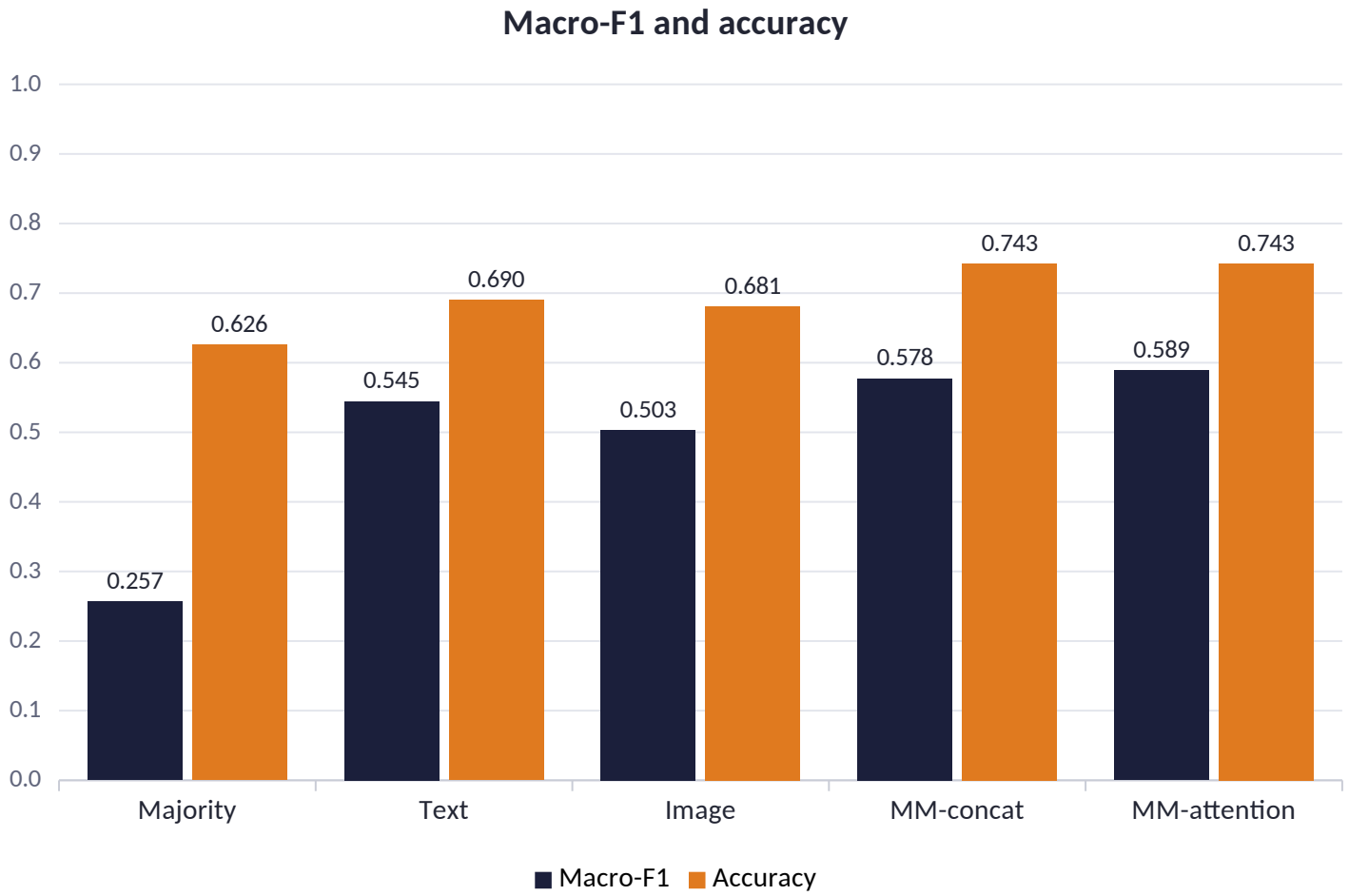
Identical for every experiment; test set never used for selection

Setting	Value
Loss	Cross-entropy, √-inverse-frequency class weights
Optimiser	AdamW · lr 2e-5 / 1e-4 / 1e-3
Schedule	10 % warm-up, linear decay
Batch · epochs	16 · ≤ 8, early stop (patience 3)
Selection	best validation macro-F1
Hardware	Apple M1, 8 GB, PyTorch MPS

- **Primary metric:** macro-F1 — an all-negative model gets ~63 % accuracy but 0.26 macro-F1
- **Uncertainty:** 95 % bootstrap CIs; paired bootstrap for model differences
- **Modality ablation:** neutral image / removed caption at test time
- **Robustness:** second split with whole source pages held out

Results: test set, random group split

874 test memes · identical for every model · 95 % CIs in notes



Model	Macro-F1	95 % CI
Majority	0.257	0.25–0.26
Text	0.545	0.50–0.59
Image	0.503	0.47–0.54
MM-concat	0.578	0.53–0.62
MM-attention	**0.589**	**0.55–0.63**

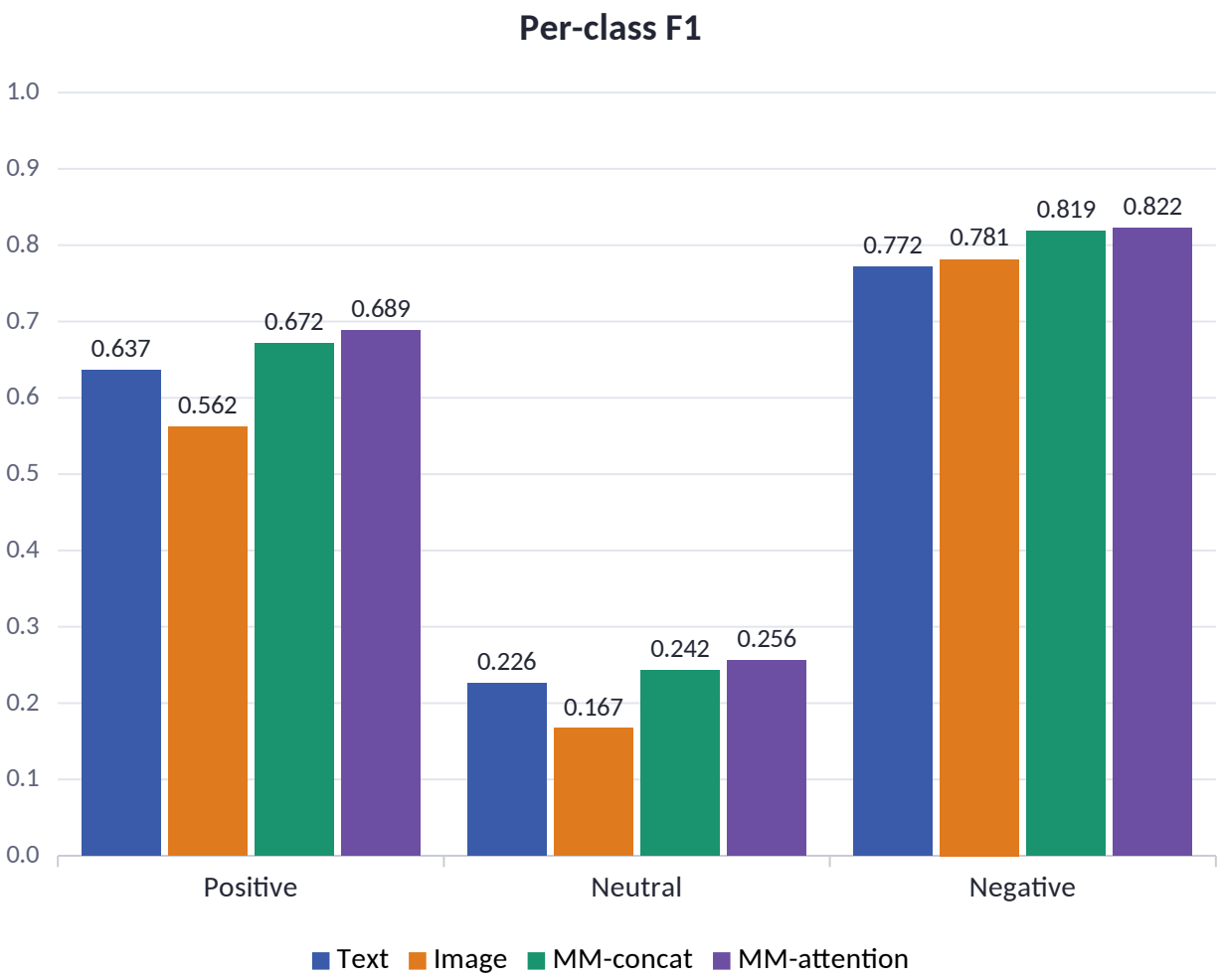
MM-concat vs Text: Δ +0.033 → not significantly different

MM-concat vs Image: Δ +0.074 → significantly better

MM-attention vs Text: Δ +0.044 → significantly better

MM-attention vs Image: Δ +0.085 → significantly better

Per-class F1 and modality ablation



Model · condition	Macro-F1	Changed
MM-concat · full input	0.578	–
MM-concat · neutral image	0.511	23%
MM-concat · no caption	0.470	14%
MM-attention · full input	0.589	–
MM-attention · neutral image	0.467	21%
MM-attention · no caption	0.376	27%

“Changed” = share of test predictions that flip when one input is neutralised. Large flips for both inputs mean the fused model relies on both.

Shortcut: the source page predicts the label

Memes from the same page share watermarks and style — and very different label mixes

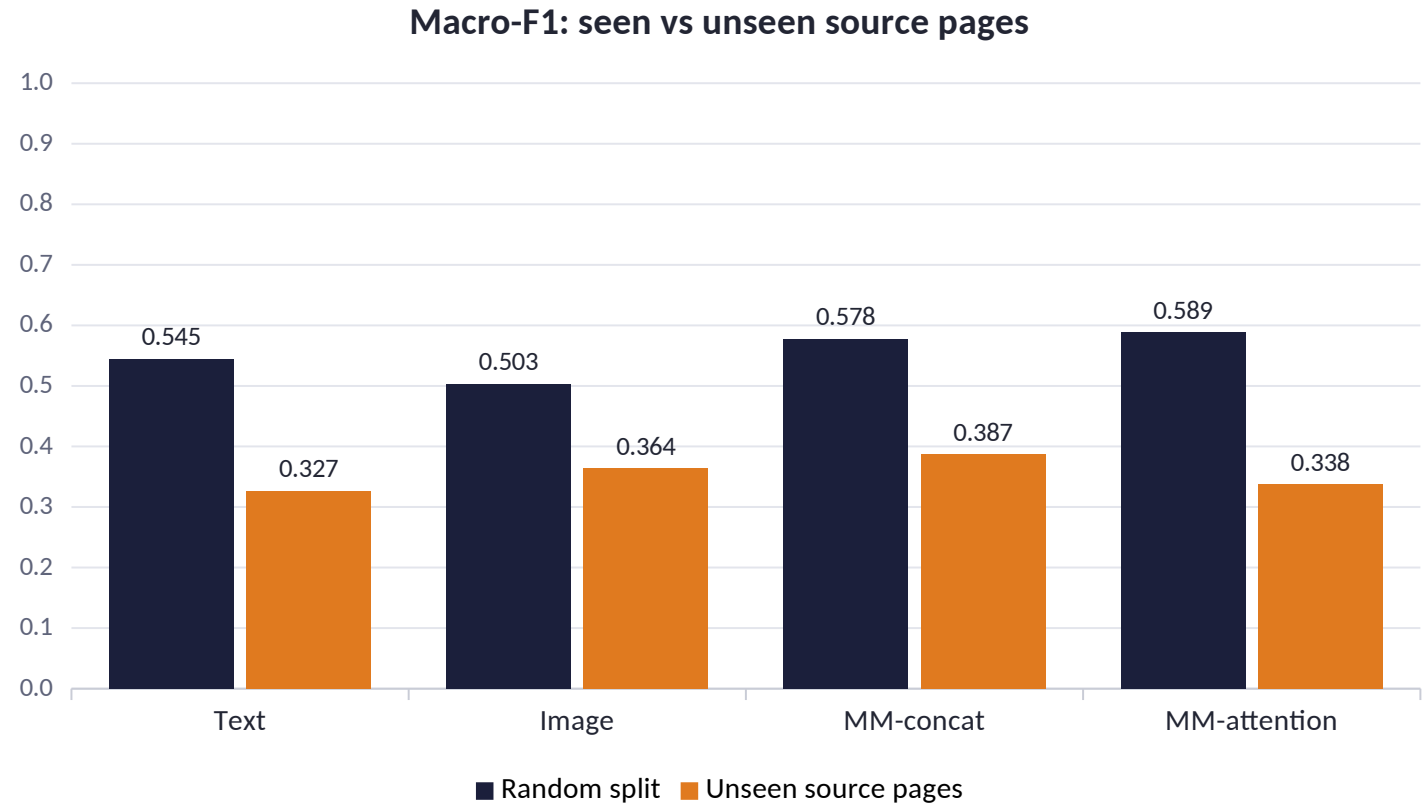
78%

test accuracy from page identity alone (no content)

0.524

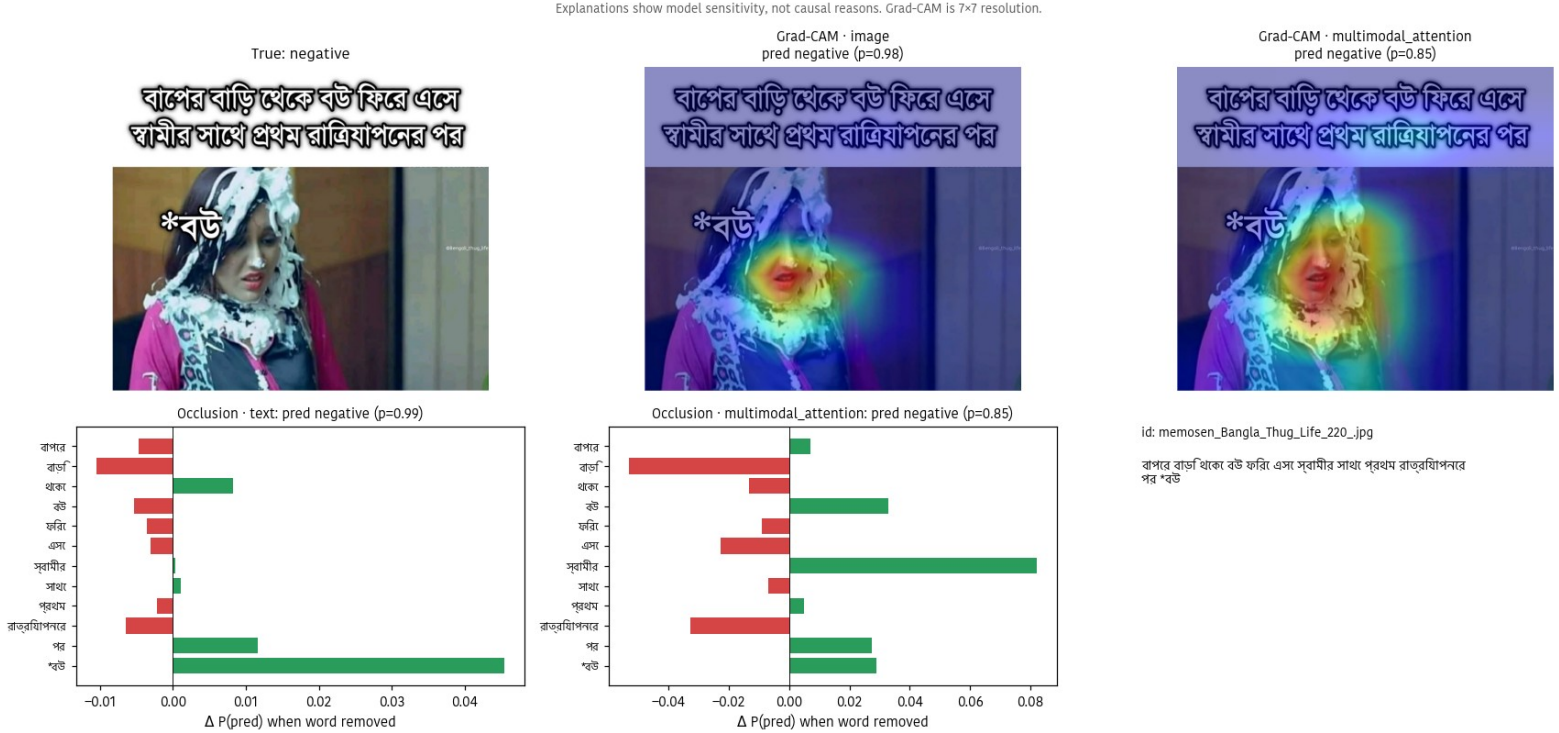
macro-F1 of that shortcut

e.g. page “KAM” is 80 % positive; “chosha” is 98.5 % negative



Different test sets (874 vs 938 memes): compare the drop per model, not absolute values across splits.

Explainability: what the models are sensitive to



- **Occlusion:** drop in P(class) when a word is removed — exact, but ignores interactions
- **Integrated Gradients:** completeness checked; hidden when it fails to converge on XLM-R
- **Grad-CAM:** 7×7 sensitivity map; often highlights caption text in the image
- **Caveat:** sensitivity ≠ cause; not proof of what an annotator meant

The IndicSent-M application

Streamlit UI + FastAPI service, CPU-compatible, Docker image

Home

objectives, languages, architecture

Analyze

text / image / multimodal input, probabilities, word highlights, Grad-CAM

Model Comparison

tables, CIs, confusion matrices, ablations, both split protocols

Dataset & Methodology

statistics, pipeline, limitations

About

motivation, technologies, future work

- **Never fakes a prediction:** no checkpoint → setup steps; optional preview is labelled DEMO
- **Confidence wording:** predicted-class probability, not a guarantee
- **Robust inputs:** corrupt / oversized images, empty text, unsupported scripts handled
- **Tested:** pytest suite covers data, models, inference, explainers, API

Conclusions

- Random split: best macro-F1 MM-attention (0.589).
- MM-concat vs text-only: not significantly different (Δ +0.033, one-sided p = 0.097; 9 comparisons, so treat p > 0.006 as fragile).
- MM-attention vs text-only: significantly better (Δ +0.044, one-sided p = 0.015; 9 comparisons, so treat p > 0.006 as fragile).
- Cross-attention vs concatenation: not significantly different.
- Page identity alone gives 78% accuracy: random-split scores mix sentiment with page recognition.
- On unseen pages every model drops 0.14–0.25 macro-F1; best is MM-concat (0.387).
- Unseen pages: cross-attention vs concat significantly worse (Δ -0.049).
- Neutral memes (~7 %) remain the hardest class for every model.

Future work

- Hinglish memes (Memotion 3, registration)
- OCR for captions
- CLIP-style vision-language encoders
- Several seeds + calibration
- Sarcasm-aware training
- Human study of explanations